

# 34721/34722 Linear Control Design 1

## Spring 2023

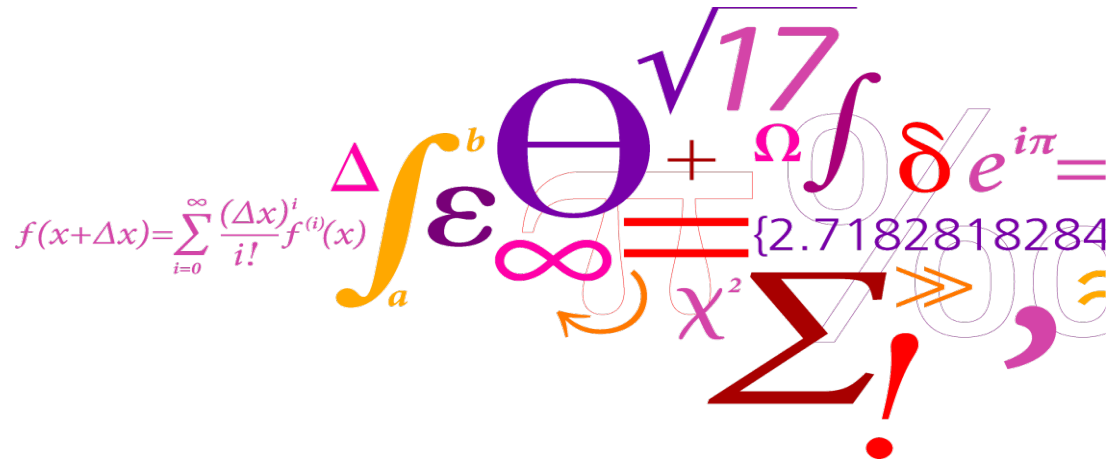
*Silvia Tolu - Dimitrios Papageorgiou*



### Lecture 13/13

**Schedule (13:00-15:00)**

- **Feed-forward control**



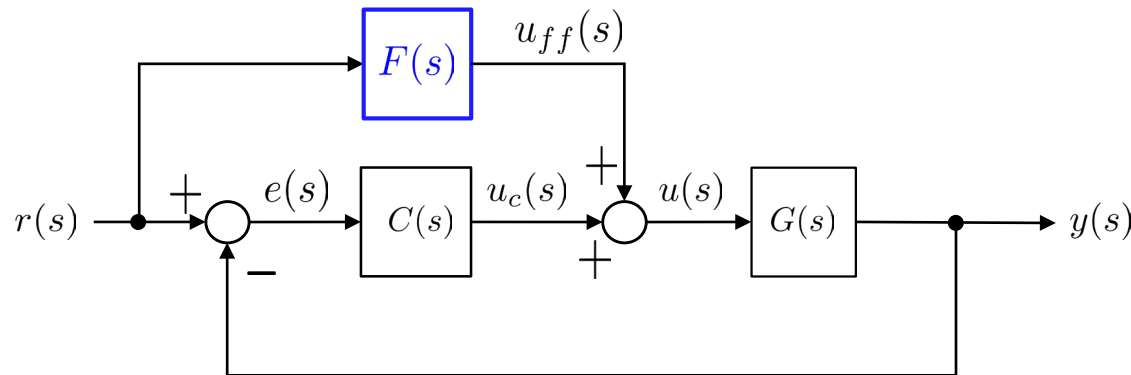
# What you will learn today

After this lecture you will be able to

- Design feed-forward controllers for improving tracking performance.
- Design feed-forward controllers for disturbance rejection.

# Feed-forward control

- A feed-forward part uses only the input (reference or disturbance) and NOT the output.
- It aims to doing the “heavy job” of the control task
  - If we want to go “from 0 to 10.45”, the feed-forward will take us to “10” and then the feedback controller will adjust the rest.
  - Any feedback-related issues (measurement noise, phase lag etc.) do not affect the performance.



# Feed-forward control

- 2 ways from  $r$  to  $y \rightarrow$  2 forward branches:  $G_{fb,1}(s) = C(s)G(s)$ ,  $G_{fb,2}(s) = F(s)G(s)$
- The (open) loop transfer function is the same:  $G_{ol}(s) = C(s)G(s)$

From  $r$  to  $y$

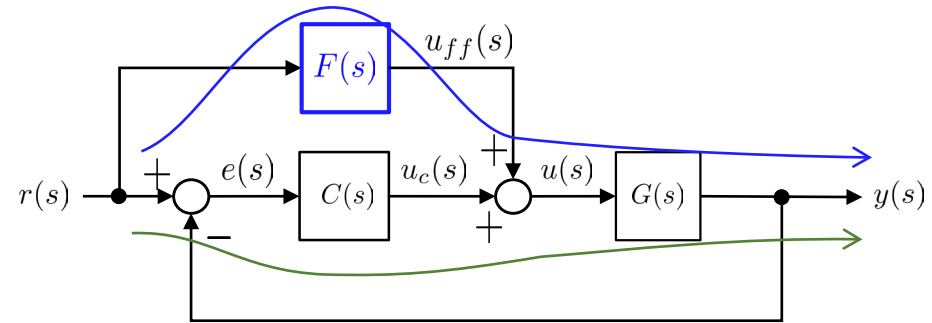
$$y(s) = G(s) \left( C(s) \underbrace{(r(s) - y(s))}_{e(s)} + F(s)r(s) \right) \Leftrightarrow$$

$$\left( 1 + \underbrace{G(s)C(s)}_{G_{ol}(s)} \right) y(s) = (G(s)C(s) + G(s)F(s)) r(s) \Rightarrow$$

$$G_{cl}(s) = \frac{F(s)G(s)}{1 + G_{ol}(s)} + \frac{G_{ol}(s)}{1 + G_{ol}(s)} = \frac{F(s)G(s) + G_{ol}(s)}{1 + G_{ol}(s)}$$

From  $r$  to  $e$

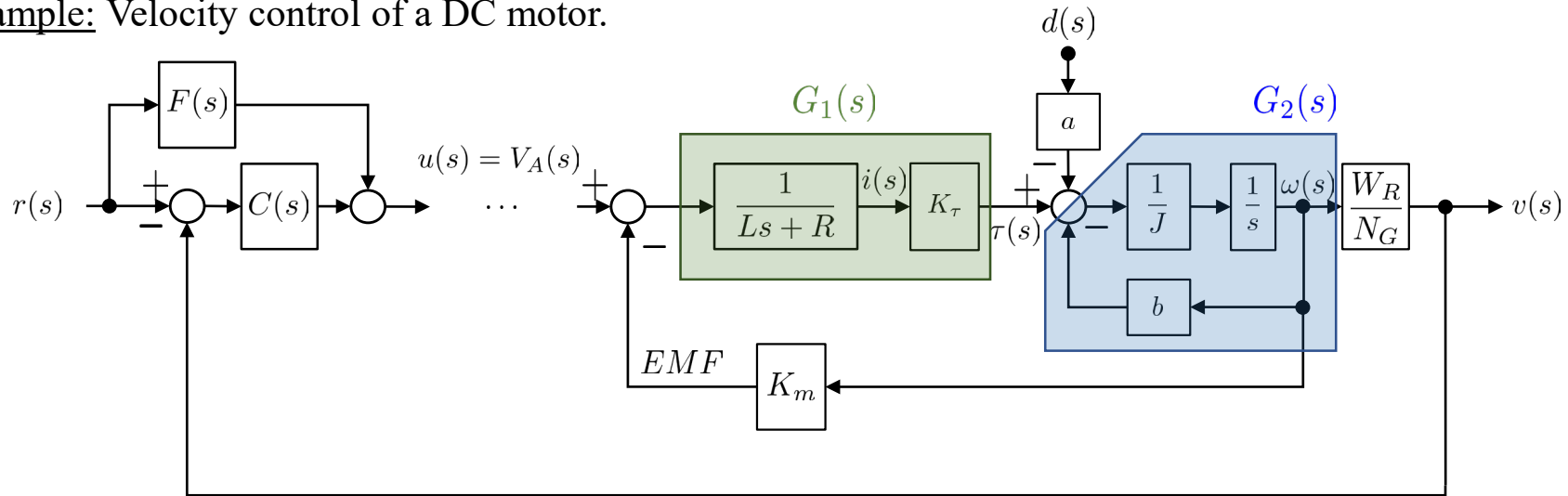
$$G_{er}(s) = \frac{-F(s)G(s)}{1 + G_{ol}(s)} + \frac{1}{1 + G_{ol}(s)} = \frac{1 - F(s)G(s)}{1 + G_{ol}(s)}$$



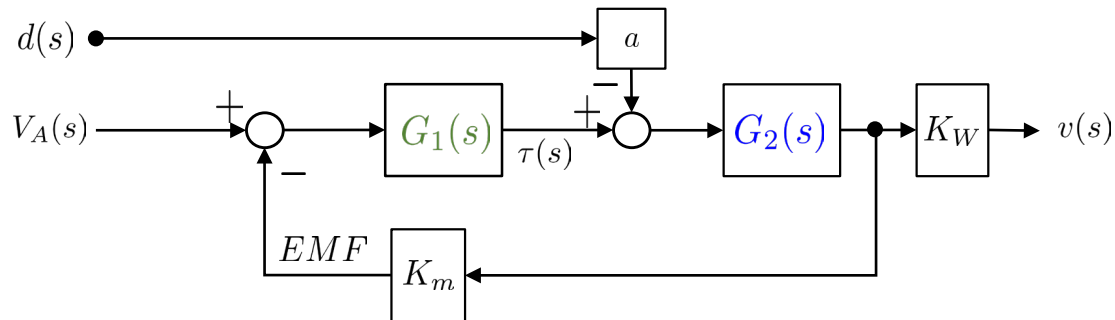
$$F(s) = \frac{1}{G(s)} \Rightarrow G_{cl}(s) = 1, G_{er}(s) = 0$$

# Feed-forward control

Example: Velocity control of a DC motor.



- First, some tidying up...



$$G_1(s) = \frac{K_\tau}{Ls + R}$$

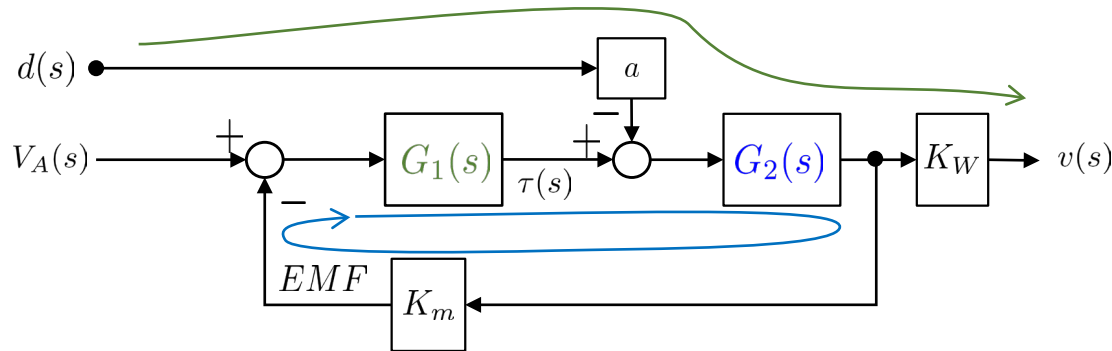
$$G_2(s) = \frac{1}{Js + b}$$

$$K_W \triangleq \frac{W_R}{N_G}$$

# Feed-forward control

Example: Velocity control of a DC motor.

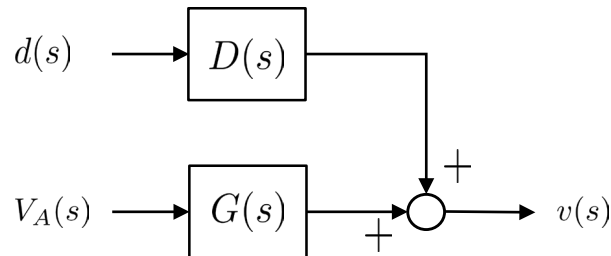
- First, some tidying up...



$$G_1(s) = \frac{K_\tau}{Ls + R}$$

$$G_2(s) = \frac{1}{Js + b}$$

$$K_W \triangleq \frac{W_R}{N_G}$$

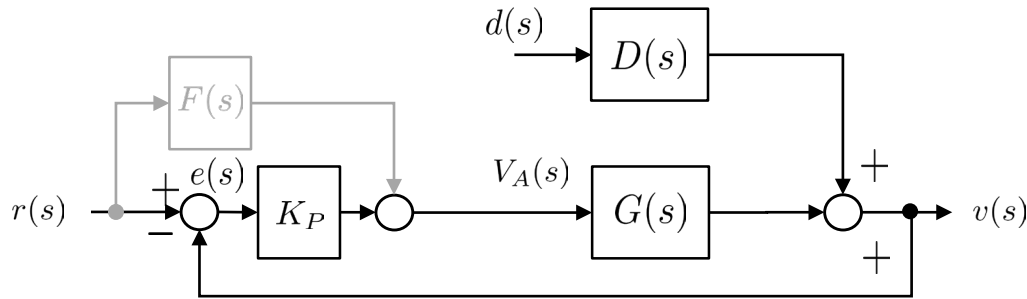


$$G(s) = K_W \frac{G_1(s)G_2(s)}{1 + K_m G_1(s)G_2(s)} = \frac{K_W K_\tau}{LJs^2 + (Lb + RJ)s + Rb + K_m K_\tau}$$

$$D(s) = -aK_W \frac{G_2(s)}{1 + K_m G_1(s)G_2(s)} = \frac{-aK_W(Ls + R)}{LJs^2 + (Lb + RJ)s + Rb + K_m K_\tau}$$

# Feed-forward control

Example: Velocity control of a DC motor.



Try only a P controller with required phase margin  $\gamma_M = 80^\circ$

$$G(s) = \frac{72}{s^2 + 41.25s + 306}, \quad D(s) = -\frac{1.012s + 40.5}{s^2 + 41.25s + 306}$$

$$\angle G(\omega_c j) = -180^\circ - \gamma_M = -100^\circ \Rightarrow \omega_c \cong 21.55 \text{ rad/s}$$

$$K_P = \frac{1}{|G(j\omega_c)|} \cong 12.542 \text{ (or 21.97 dB)}$$

$$L = 0.05 \text{ H}$$

$$R = 2 \Omega$$

$$K_\tau = 0.016 \text{ Nm/A}$$

$$K_m = 0.016 \text{ Vs}$$

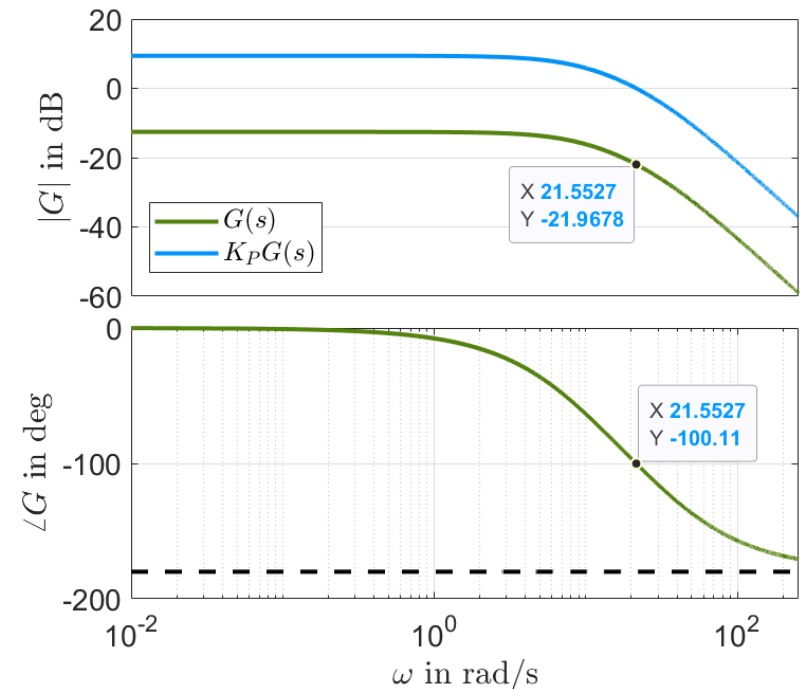
$$J = 20 \cdot 10^{-6} \text{ Nms}^2$$

$$b = 25 \cdot 10^{-6} \text{ Nms}$$

$$W_R = 0.045 \text{ m}$$

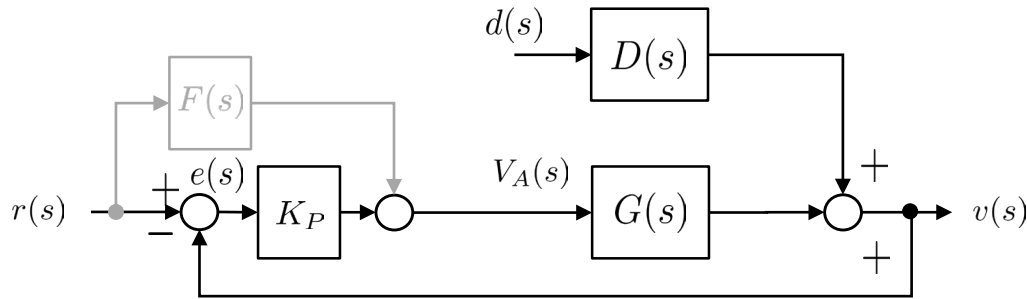
$$N_G = 10$$

$$a = W_R/N_G$$



# Feed-forward control

Example: Velocity control of a DC motor.



$$G(s) = \frac{72}{s^2 + 41.25s + 306}, \quad K_P = 12.542$$

$$G_{cl}(s) = \frac{K_P G(s)}{1 + K_P G(s)} = \frac{903}{s^2 + 41.25s + 1209}$$

$$G_{er}(s) = \frac{1}{1 + K_P G(s)} = \frac{s^2 + 41.25s + 306}{s^2 + 41.25s + 1209}$$

$$e_{ss} = \lim_{s \rightarrow 0} s G_e(s) \frac{1}{s} = \frac{1}{1 + K_P G(0)} = \frac{306}{1209} = 0.2531$$

$$L = 0.05 \text{ H}$$

$$R = 2 \Omega$$

$$K_\tau = 0.016 \text{ Nm/A}$$

$$K_m = 0.016 \text{ Vs}$$

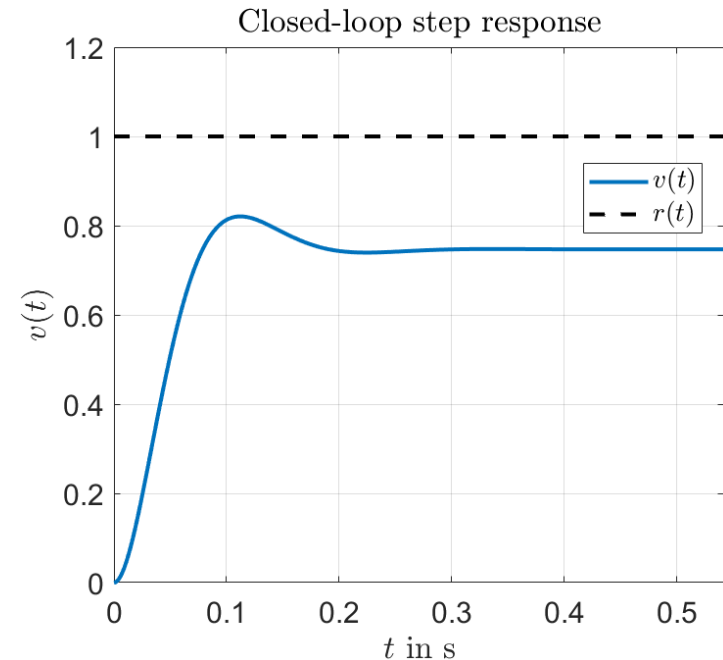
$$J = 20 \cdot 10^{-6} \text{ Nms}^2$$

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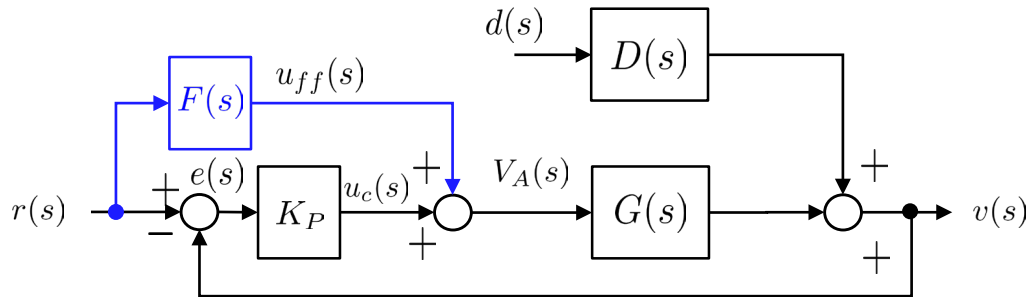
$$a = W_R / N_G$$





# Feed-forward control

Example: Velocity control of a DC motor.



$$G_{er}(s) = \frac{1}{1 + K_P G(s)} - \frac{F(s)G(s)}{1 + K_P G(s)} = \frac{1 - F(s)G(s)}{1 + K_P G(s)}$$

For  $F(s) = \frac{1}{G(s)}$  we have zero error:  $G_{er}(s) = 0$

$$F(s) = \frac{1}{G(s)} = \frac{s^2 + 41.25s + 306}{72}$$

Can we work with this?

$$L = 0.05 \text{ H}$$

$$J = 20 \cdot 10^{-6} \text{ Nms}^2$$

$$R = 2 \Omega$$

$$b = 25 \cdot 10^{-6} \text{ Nms}$$

$$K_\tau = 0.016 \text{ Nm/A}$$

$$W_R = 0.045 \text{ m}$$

$$K_m = 0.016 \text{ Vs}$$

$$N_G = 10$$

$$a = W_R/N_G$$

$$G(s) = \frac{72}{s^2 + 41.25s + 306}, \quad K_P = 12.542$$

- Improper transfer function

$$u_{ff}(s) = \frac{s^2 + 41.25s + 306}{72} r(s) \Rightarrow$$

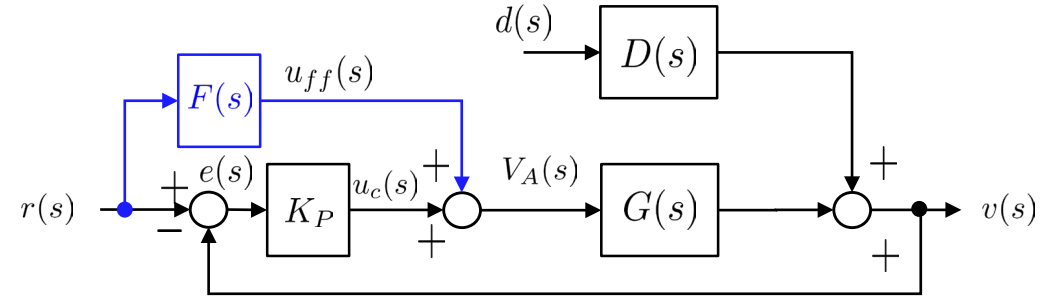
$$u_{ff}(t) = \frac{1}{72} (\ddot{r}(t) + 41.25\dot{r}(t) + 306r(t))$$

- Requires derivatives of reference signal
- Cannot be implemented in Simulink as a transfer function block.

# Feed-forward control

Example: Velocity control of a DC motor.

$$G_{er}(s) = \frac{1 - F(s)G(s)}{1 + K_P G(s)}, \quad F(s) = \frac{1}{G(s)} \Leftrightarrow G_{er}(s) = 0$$



$$G(s) = \frac{72}{s^2 + 41.25s + 306}, \quad K_P = 12.542$$

- $F(s) = \frac{1}{G(0)} = \frac{306}{72} \triangleq F_0$  (emphasis on steady state)

- Use a low-pass filter with DC gain 1 and as many poles as “missing”  $1/G(s)$

$$F(s) = \frac{1}{G(s)} G_{filt}(s) = \frac{s^2 + 41.25s + 306}{72} \cdot \frac{1}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

$$F(s) = \frac{s^2 + 41.25s + 306}{72(0.0021s + 1)^2} \quad (\text{emphasis on a frequency range})$$

- We need to have the same or higher order as the numerator.
- Filter poles must be stable.
- Choose poles ~10 times faster than the fastest pole of  $G(s)$  – here we chose  $\tau_1 = \tau_2 = 0.0021$  sec.

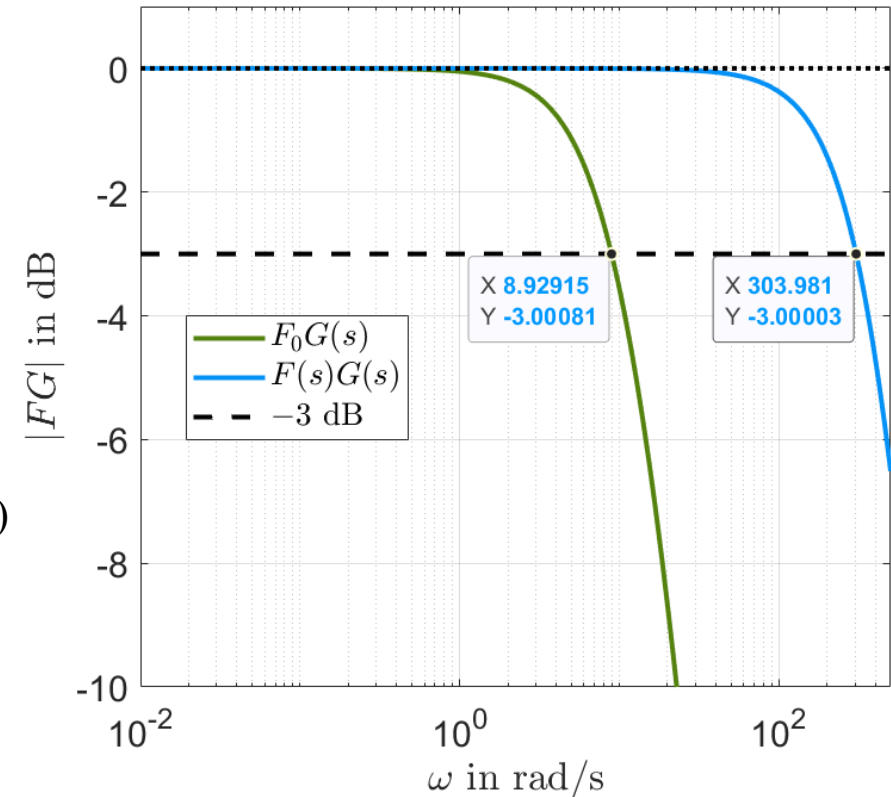
# Feed-forward control

Example: Velocity control of a DC motor.

$$G_{er}(s) = \frac{1 - F(s)G(s)}{1 + K_P G(s)}, \quad F(s) = \frac{1}{G(s)} \Leftrightarrow G_{er}(s) = 0$$

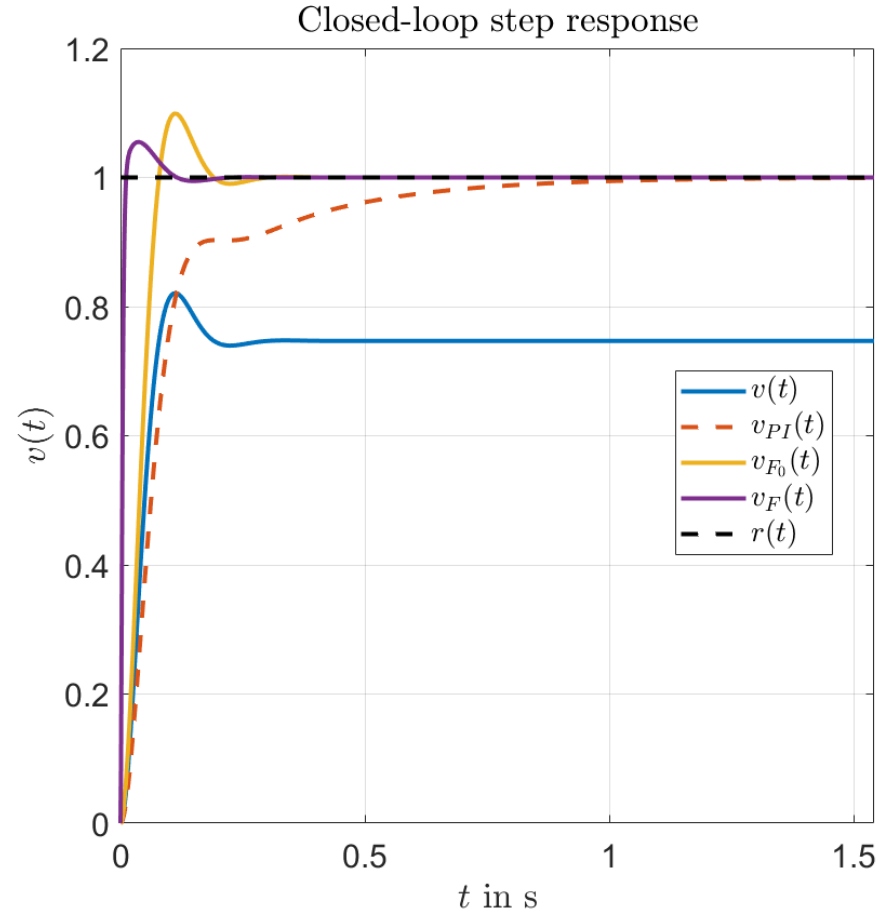
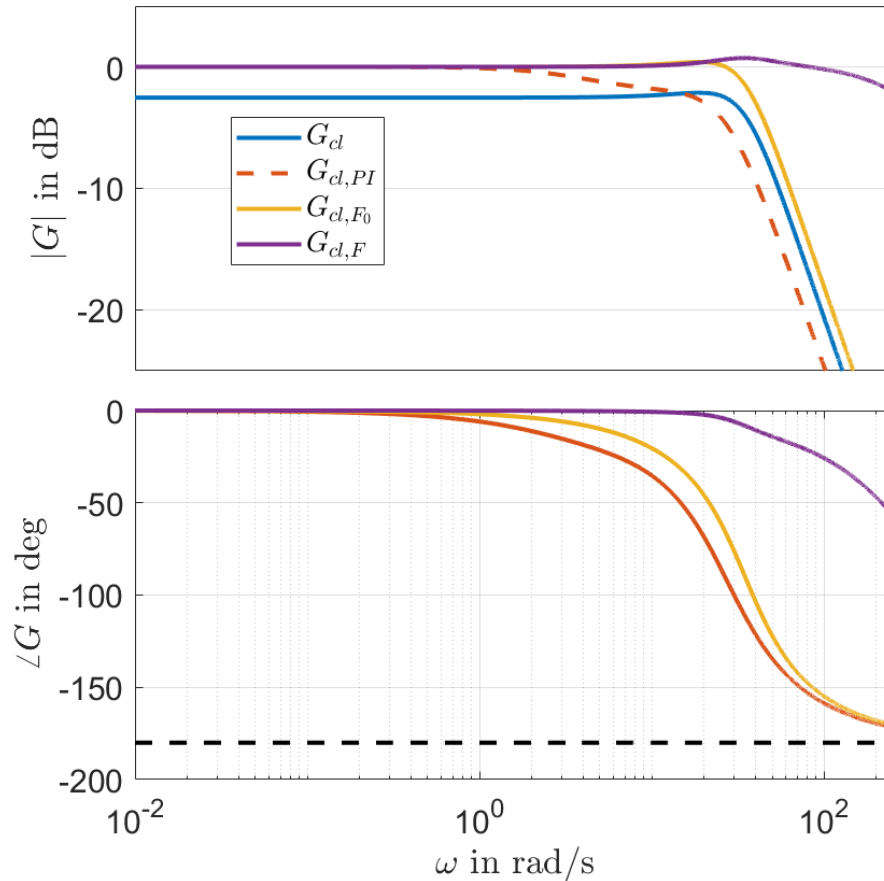
Requires reference derivatives. If not available:

- $F(s) = \frac{1}{G(0)} = \frac{306}{72} \triangleq F_0$  (emphasis on steady state)
- $F(s) = \frac{s^2 + 41.25s + 306}{72(0.0021s + 1)^2}$  (emphasis on a frequency range)



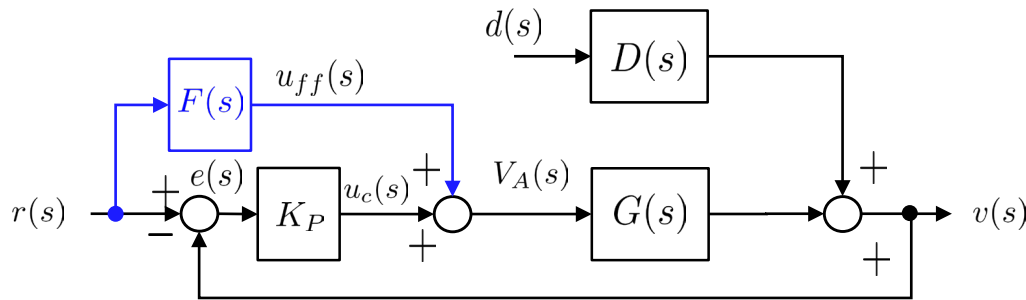
# Feed-forward control

Example: Velocity control of a DC motor.



# Feed-forward control

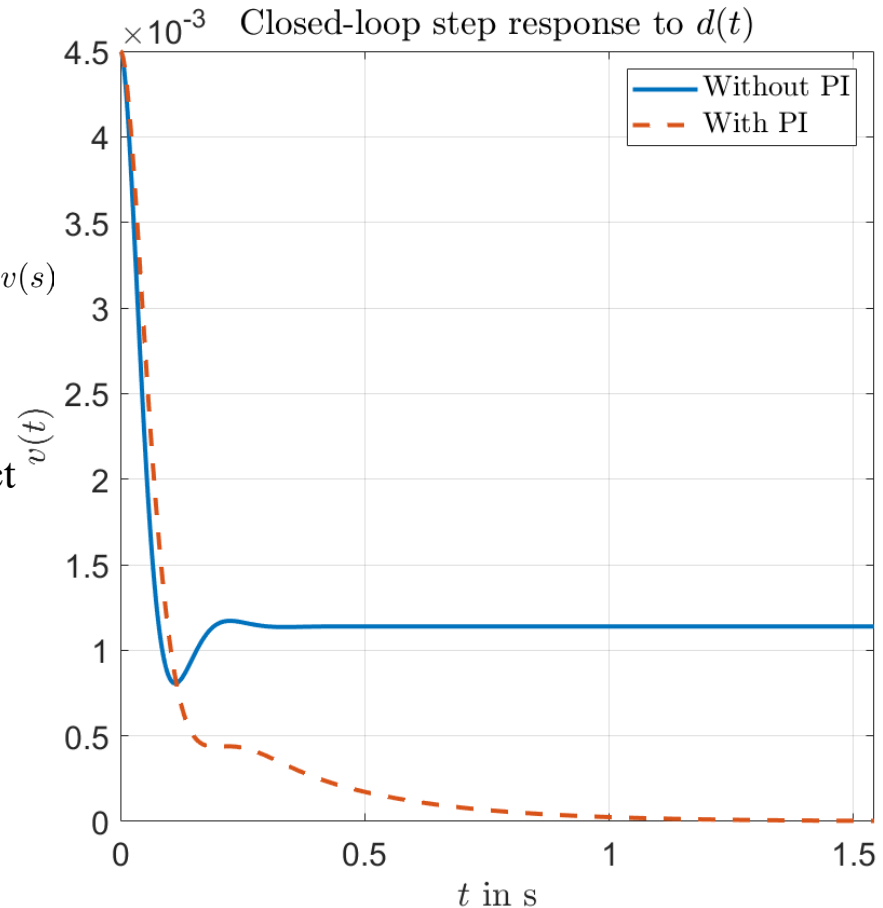
Example: Velocity control of a DC motor. What about the disturbance?



- The feed-forward from the reference does not affect the disturbance.

$$G_{vd}(s) = \frac{D(s)}{1 + K_P G(s)}$$

FF\_example.m



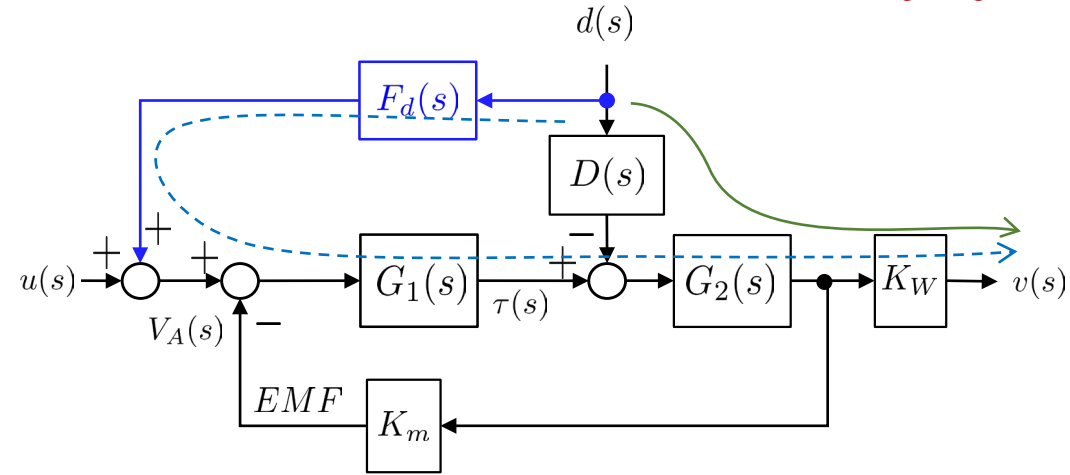
# Feed-forward control

## Comprehension questions

- Is the feed-forward control affecting stability?
- Does the feed-forward control amplify the measurement noise?
- Does the feed-forward control contribute to integrator wind-up issues?
- Does the feed-forward control (from the reference) helps with disturbance rejection?
- If the system is type-1, is a “static” feed-forward controller ( $F(s) = F_0$ ) useful?

# Feed-forward control – disturbance rejection

- The feed-forward from the reference does not affect the disturbance.
- If we know or have an estimate of the disturbance, we can use it as feed-forward term.
- 2 ways from  $d$  to  $v \rightarrow 2$  forward branches:



$$G_{vd}(s) = \frac{-K_W D(s) G_2(s)}{1 + K_m G_1(s) G_2(s)} + \frac{K_W F_d(s) G_2(s) G_1(s)}{1 + K_m G_1(s) G_2(s)}$$

$$= \frac{K_W G_2(s)}{1 + K_m G_1(s) G_2(s)} (F_d(s) G_1(s) - D(s))$$

- Total disturbance rejection with  $F_d(s) = \frac{D(s)}{G_1(s)}$ .

$$G_1(s) = \frac{0.016}{0.05s + 2}, D(s) = 0.02 \Rightarrow F_d(s) = 0.02 \frac{0.05s + 2}{0.016}$$

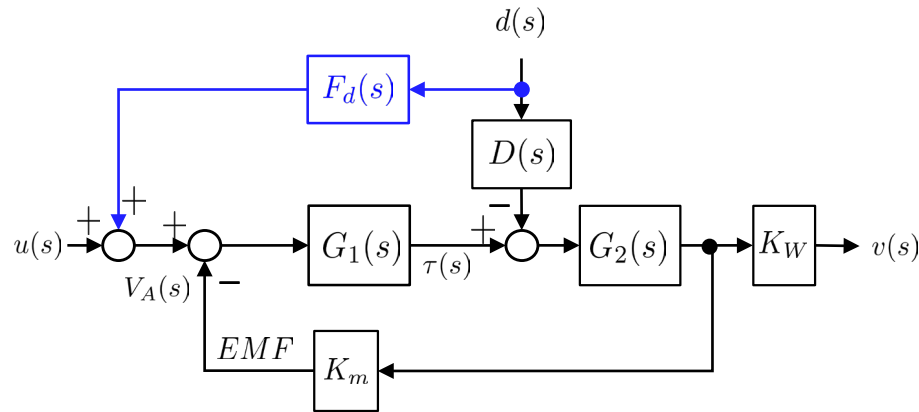
**Not so fast:** More zeros than poles!

- Option 1:** Consider only a static feed-forward.  

$$F_d(s) = \frac{D(0)}{G_1(0)} = 0.02 \frac{2}{0.016} = 2.5 \triangleq F_{d,0}$$
- Option 2:** Add some stable poles (faster than the fastest pole of  $G_1(s)$ ) to get the same or higher order as the numerator – here we chose  $\tau = 0.005$  sec.

$$F_d(s) = \frac{D(s)}{G_1(s)} G_{filt}(s) = 0.02 \frac{(0.05s + 2) \cdot 1}{0.016(0.005s + 1)}$$

# Feed-forward control – disturbance rejection

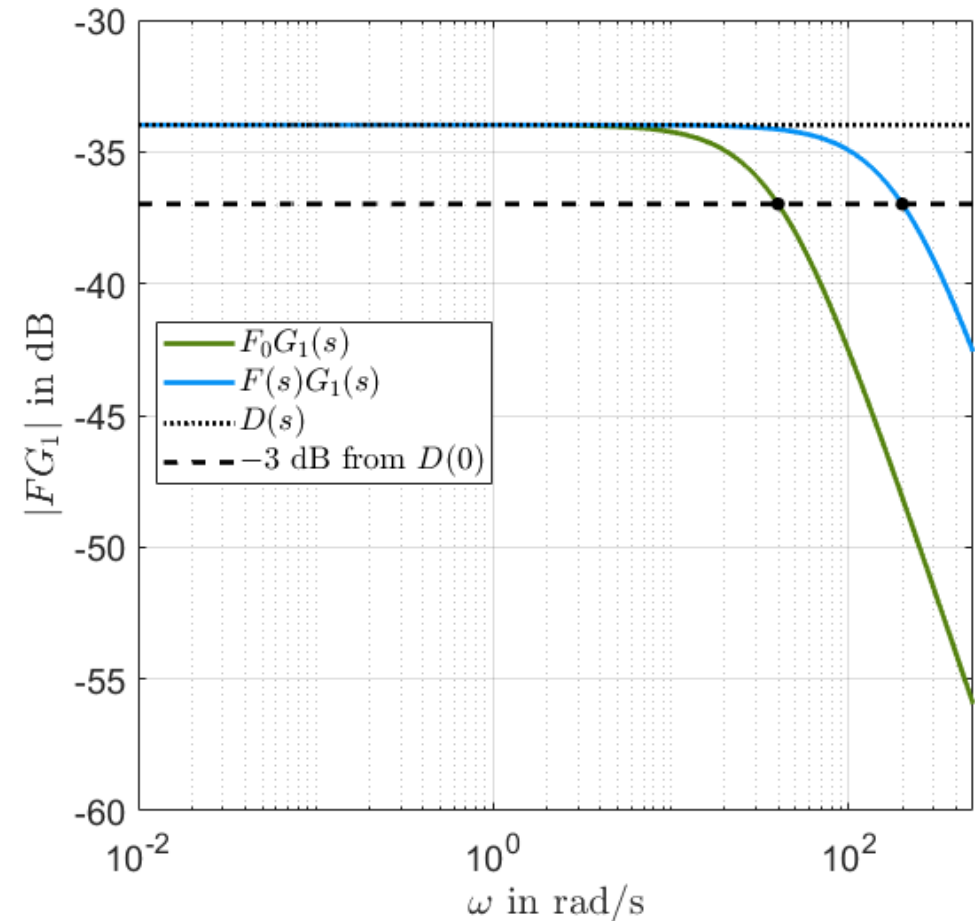


- **Option 1:** Consider only a static feed-forward.

$$F_d(s) = \frac{D(0)}{G_1(0)} = 0.02 \frac{2}{0.016} = 2.5 \triangleq F_{d,0}$$

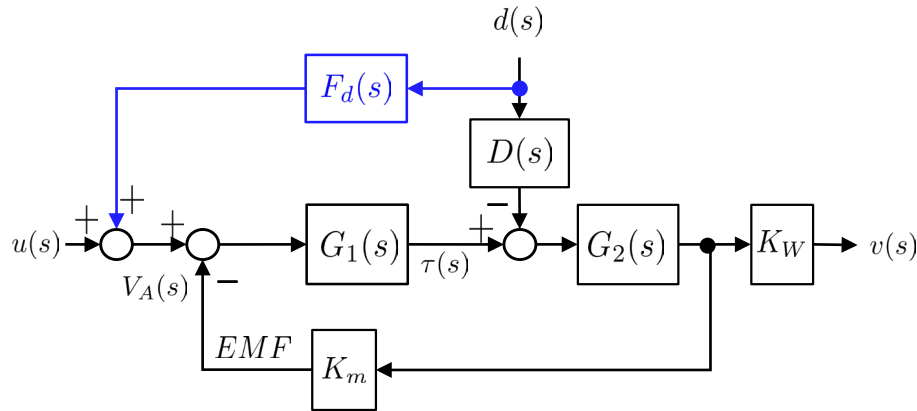
- **Option 2:** Add some stable poles (faster than the fastest pole of  $G_1(s)$ ) to get the same or higher order as the numerator – here we chose  $\tau = 0.005$  sec.

$$F_d(s) = \frac{D(s)}{G_1(s)} G_{filt}(s) = 0.02 \frac{(0.05s + 2) \cdot 1}{0.016(0.005s + 1)}$$





# Feed-forward control – disturbance rejection



- Option 1

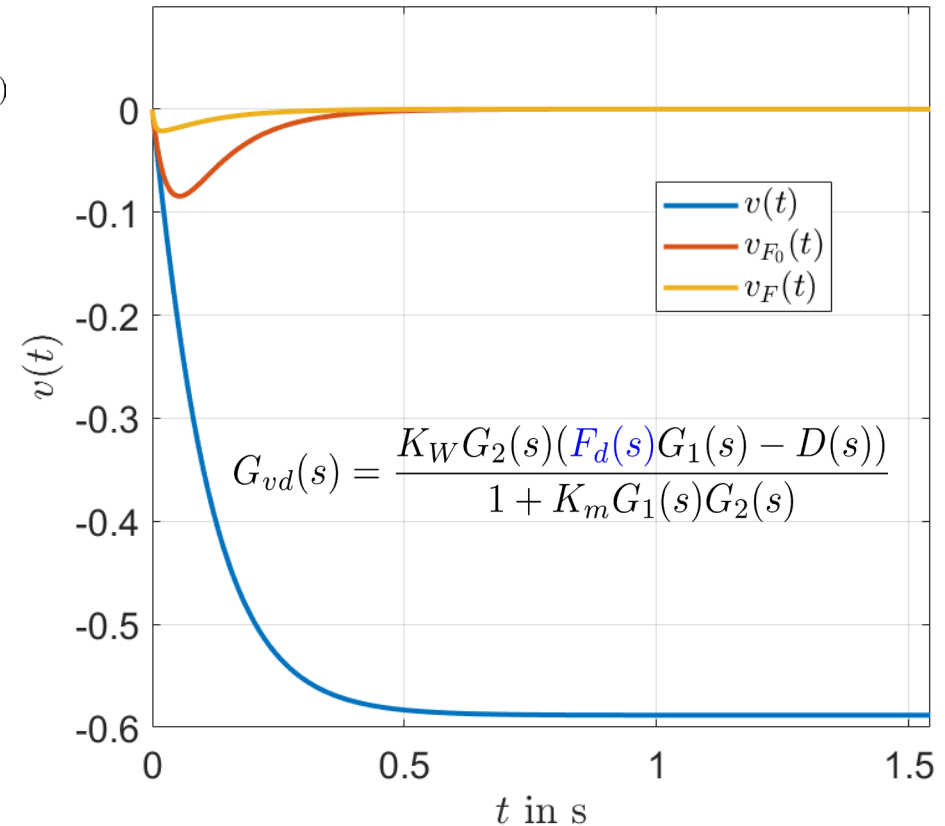
$$F_d(s) = \frac{D(0)}{G_1(0)} = 0.02 \frac{2}{0.016} = 2.5 \triangleq F_{d,0}$$

- Option 2

$$F_d(s) = \frac{D(s)}{G_1(s)} G_{filt}(s) = 0.02 \frac{(0.05s + 2) \cdot 1}{0.016(0.005s + 1)}$$

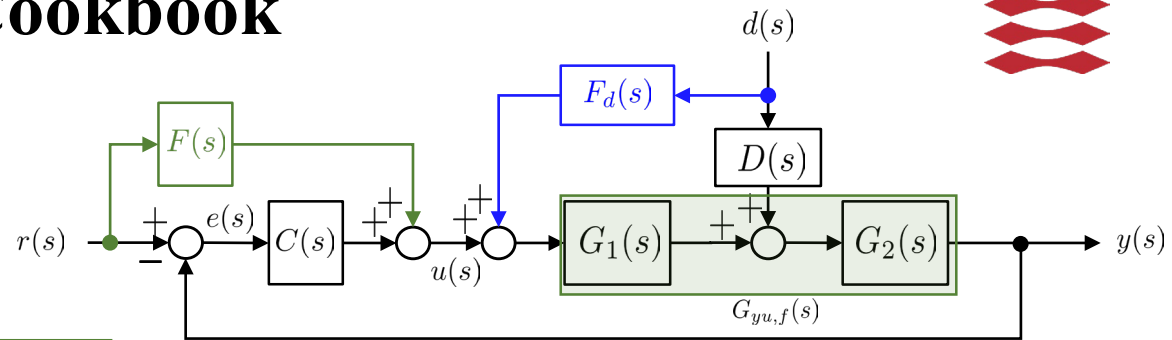
FF\_disturbance\_example.m

Closed-loop disturbance step response



# Feed-forward control – Cookbook

$$F_d(s) = \frac{D(s)}{G_1(s)} G_{filt}(s) = 0.02 \frac{(0.05s + 2) \cdot 1}{0.016(0.005s + 1)}$$



## For the reference feed-forward control

1. Find the forward branch transfer function between the control input  $u$  and the output  $y$ .

$$G_{yu,f}(s) = G_1(s)G_2(s)$$

2. Design the feed-forward transfer function as

$$F(s) = \frac{1}{G_{yu,f}(0)}$$

3. If there is not much model uncertainty and  $G_{yu,f}(s)$  has no zeros on the RHP design  $F(s)$  as

$$F(s) = \frac{G_{filt}(s)}{G_{yu,f}(s)}, \quad G_{filt}(s) = \frac{1}{(\tau_f s + 1)^n}$$

where  $n$  is the (number of poles) – (number of zeros) in  $G_{yu,f}(s)$ . The filter pole  $1/\tau_f$  is 5-10 times faster than the fastest pole of  $G_{yu,f}(s)$ . If  $n < 0$ , set  $n = 0$ .

## For the disturbance feed-forward control

1. Find the forward branch transfer function between the control input  $u$  and the summation point where the disturbance enters the system ( $G_I(s)$ ).

2. Design the feed-forward transfer function as

$$F(s) = \frac{D(0)}{G_1(0)}$$

3. If there is not much model uncertainty and  $G_I(s)$  has no zeros on the RHP design  $F(s)$  as

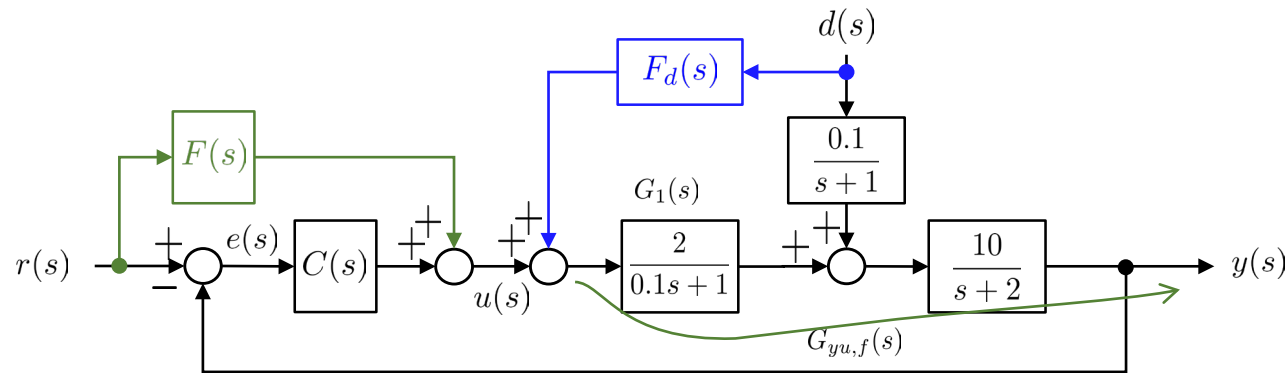
$$F(s) = \frac{D(s)G_{filt}(s)}{G_1(s)}, \quad G_{filt}(s) = \frac{1}{(\tau_f s + 1)^n}$$

where  $n$  is the (number of poles) – (number of zeros) in  $G_I(s)/D(s)$ . The filter pole  $1/\tau_f$  is 5-10 times faster than the fastest pole of  $G_I(s)$ . If  $n < 0$ , set  $n = 0$ .

# Feed-forward control

## Exercise 1

For the system in the given block diagram

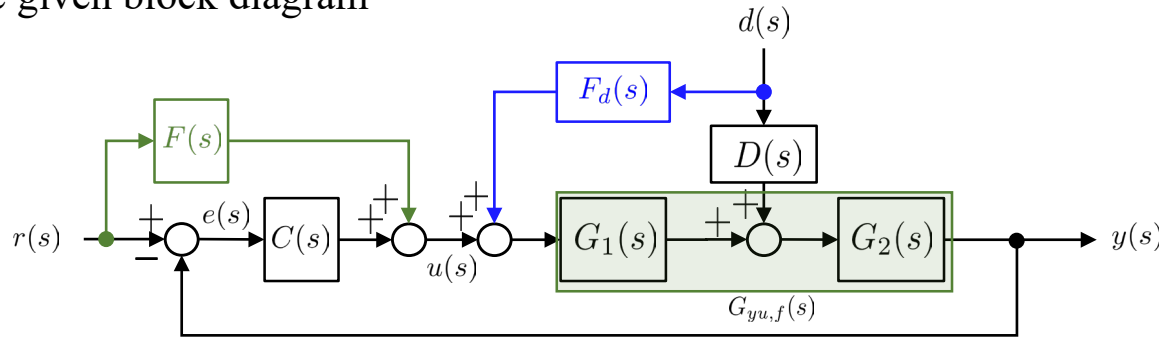


- Design  $F(s)$  and  $F_d(s)$  as static gains.
- Design  $F(s)$  and  $F_d(s)$  to work in a frequency range. What is this range for each feed-forward controller?

# Feed-forward control

## Exercise 2

For the system in the given block diagram



- If  $D(s) = 0.1$  and  $G_I(s)$  is given by  $G_1(s) = \frac{s-2}{(s+1)(s^2+3s+1)}$ , what is a good choice for  $F_d(s)$ ?
- If  $G_2(s)$  or  $G_I(s)$  have complex poles, how can we determine the feed-forward filter frequency?
- If  $G_{yu,f}(s)$  is given by  $G_{yu,f}(s) = \frac{s^2 + 0.0002s + 49}{(s+1)(s^2 + 14s + 100)}$  with some uncertainty, is it a good idea to use a non-static feed-forward controller?

# Today you learned

- How to design feed-forward controllers for improving tracking performance.
- How to design feed-forward controllers for disturbance rejection.

## Today's Exercises

Group work: Continue on REGBOT balance (mandatory assignment)

Home alone: Continue on the theoretical exercises LCD1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

# So how does one design controllers?

## Pre-design

- Study the system, i.e. its properties, limitations. Look at previous work, literature and datasheets.
- Define a reasonable ideal behaviour and a control objective (remember the example with the container ship → it cannot move 1 km in 1 sec).
- Translate performance specifications into frequency domain requirements (bandwidth, phase margin etc.).

## Modelling

- Get a coarse model on the paper if possible. Use physics and math.
- Run A LOT of open-loop experiments to find parameters. Linearise if the system is nonlinear. Always pick a linearization point that makes sense for the operation of the system.
- Alternatively, use the input/output data to get a transfer function. Try different number of poles/zeros.
- Run extra experiments for validation of the model to avoid overfitting.

## Control design

- Decide on which control parts you will include (based on the specifications, steady-state error etc.).
- Pick a crossover frequency based on the specifications → find the controller transfer function.
- Implement and simulate the closed-loop system. Re-iterate the design until the performance is good.

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## [34765](#) **Digital control** (5 ETCS, fall semester)

» All you need to know in order to implement your control designs in real life



## [34745](#) **Linear control design 2** (10 ETCS, fall semester)

» The absolutely necessary for control design



## [34746](#) **Robust and Fault-tolerant control** (10 ETCS, spring semester)

» Fault diagnosis, fault handling, robust control design



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» Neural control based on architectures and functional principles of the brain



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» Nonlinear control design



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» Control, planning and estimation for marine robots

